



Purpose

The purpose of the Safe Operating Procedure (SOP) is to provide employees the safe practices and identify all possible hazards for employees to safely enter & exit into tanks for cleaning and maintenance.

LOW HAZARD ATMOSPHERE REMINDER:

The following procedure is only for low hazard atmospheres only!!
The air inside the vacuum tank and or storage tank must read:

- a. 0 – LEL 0 - CO
- b. 0 - H2S 20.9 - O2

If the monitor reads anything other than clean air, entry into the vacuum tank is not permitted for any reason. Contact your supervisor for further instructions

Scope

This procedure is intended for all Silverpoint employees.

Hazard Assessment & Identification

Listed below are typical hazards expected in any general confined space entry.

It's required that; a complete site specific hazard assessment must be completed reviewed/signed off by project supervisor or designate before each project starts to identify the specific hazards for the project/site.

In British Columbia all hazard assessments must be signed off by a Professional Body.

WARNING: No one is permitted to enter and work in a Confined Space unless the following requirements have been satisfied:

- Supervisor and worker have valid H2S Alive Certificate;
- Supervisor and worker have valid Confined Space training certificate;
- An assessment of the confined space has been performed . The assessment
- must include an air quality deficient atmosphere test results must be recorded .
- The work must be performed with one worker outside the confined space, standing watch .
- Each worker must have a personal tested gas monitor.

Examples of some of the hazards are listed below may include but not limited too:

- Thermal Energy-Combustion/Fire/Explosion
- Physical Stress-Excessive Noise
- Exposure To Flammable Substance
- Mental Stress, Claustrophobia
- Exposure To Toxic/Carcinogenic Substance
- Personal Injury Slip, Trip Hazard
- Exposure To Corrosive Substance
- Equipment Damage
- Exposure To Irritating/Sensitizing Substance
- Fall from elevated work site
- Temperature Extremes-Personal
- Ergonomics
- Potential Energy-Pressurized Gases/Liquids
- Equipment Failure/Stress



- Potential Energy-Stored Energy
- Mud, Ice, Weather
- Deficient Atmosphere-Toxic Or Lack Of Oxygen
- Lifting, Twisting, Pushing, Pulling
- Physical Stress-Reduced Or Restricted Visibility • Physical Stress-Personal Injury

EHS Requirements

- Personal Protective Equipment including approved Steel Toed Boots, or Rubber Boots, Fire Resistant coveralls, eye and face protection, hard hat, gloves;
- Air monitoring equipment that is certified, and calibrated for the potential hazardous atmospheric conditions, and has been correctly bump tested before use;
- Lanyard, harness or other acceptable rescue retrieval system that allows for rescue from outside of the confined space;
- Competent confined space entry attendant or central dispatch;
- Confined Space Permit must be completed and signed off by a site Supervisor;
- Alberta OH&S
- Silverpoint Corporate Code of Practice
- Silverpoint Task List
- Silverpoint THA 4.04

Bump testing procedure for the GX-2009 only.

If you are using a different type of monitor follow the manufactures procedure.

The following must be performed in a known clean atmosphere:

- Press the power/mode button for 1 second until it beeps to turn the monitor on.
- The monitor will display the days remaining until a calibration is needed.
- The monitor will go thru its warm up cycle and perform a self-test.
- Monitor will perform an automatic fresh air adjustment, be sure you are in a fresh air.
- Monitor should display fresh air readings, 0-Co, 0-H2s, 0-LEL and 20.9 – O2.
- Place the calibration cup, hose and balloon assembly on the bottom of the monitor.
- Calibration cup only goes on 1 way, look for the 4 small round indents and match them to the monitor.
- Put the small clear tube into the bump gas nozzle.
- Kink the clear tubing before you put bump gas into the balloon.
- Slowly pull bump test gas canister trigger until a small amount of bump gas starts to flow.
- Monitor should alarm. (Beep/vibrate/flashing red light). All 4 numbers must be flashing.
- Remove calibration cup from monitor to allow fresh air to reach monitor heads.
- Press the power/mode button to go to peak reading screen and record numbers of the bump test.
- To zero peak reading press the AIR button until it tells you to release.

If there is any doubt of any type WORK MUST STOP and a Supervisor must be contacted.